

Original Article

Genome-wide investigation and expression analysis of APETALA-2 transcription factor subfamily reveals its evolution, expansion and regulatory role in abiotic stress responses in *Indica* Rice (*Oryza sativa* L. *ssp. indica*)

Sohaib Ahmed^{a,1}, Muhammad Abdul Rehman Rashid^{b,c,g,1}, Syed Adeel Zafar^d, Muhammad Tehseen Azhar^{a,e}, Muhammad Waqas^a, Muhammad Uzair^d, Iqrar Ahmad Rana^f, Farrukh Azeem^g, Gyuhwa Chung^h, Zulfiqar Aliⁱ, Rana Muhammad Atif^{a,j,*}

^a Department of Plant Breeding and Genetics, University of Agriculture Faisalabad, Faisalabad 38040, Pakistan

^b State Key Laboratory for Conservation and Utilization of Bio-Resources in Yunnan, Research Center of Perennial Rice Engineering and Technology in Yunnan, School of Agriculture, Yunnan University, Kunming 650500, China

^c Industrial Crops Research Institute, Yunnan Academy of Agricultural Sciences, Kunming 650200, China

^d National key facility for Crop Gene Resources and Genetic Improvement, Institute of Crop Science, Chinese Academy of Agricultural Sciences, Beijing, China

^e School of Agriculture Sciences, Zhengzhou University, Zhengzhou 450000, China

^f Center for Agricultural Biochemistry and Biotechnology, University of Agriculture, Faisalabad 38040, Pakistan

^g Department of Bioinformatics and Biotechnology, Government College University, Faisalabad 38000, Pakistan

^h Department of Biotechnology, Chonnam National University, Chonnam 59626, Republic of Korea

ⁱ Institute of Plant Breeding and Biotechnology, Muhammad Nawaz Shareef University of Agriculture, Multan 66000, Pakistan

^j Center for Advanced Studies in Agriculture and Food Security (CAS-AFS), University of Agriculture Faisalabad, Faisalabad-38040 Pakistan

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ABSTRACT

Rice is an important cereal crop that serves as staple food for more than half of the world population. Abiotic stresses resulting from changing climatic conditions are continuously threatening its yield and production. Genes in *APETALA-2* (*AP2*) family encode transcriptional regulators implicated during regulation of developmental processes and abiotic stress responses but their identification and characterization in *indica* rice was still missing. In this context, twenty-six genes distributed among eleven chromosomes in *Indica* rice encoding *AP2* transcription-factor subfamily were identified and their diverse haplotypes were studied. Phylogenetic analysis of *OsAP2* TF family-members grouped them into three clades indicating conservation of clades among cereals. Segmental duplications were observed to be principal route of evolution, supporting the higher positive selection-pressure, which were estimated to be originated about 10.57 to 56.72 million years ago (MYA). Conserved domain analysis and intron-exon distribution pattern of identified *OsAP2s* revealed their exclusive distribution among the specific clades of the phylogenetic tree. Moreover, the members of *osa-miR172* family were also identified potentially targeting four *OsAP2* genes. The real-time quantitative expression profiling of *OsAP2s* under heat stress conditions in contrasting *indica* rice genotypes revealed the differential expression pattern of *OsAP2s* (6 genes up-regulated and 4 genes down-regulated) in stress- and genotype-dependent manner. These findings unveiled the evolutionary pathways of *AP2*-TF in rice, and can help the functional characterization under developmental and stress responses

Abbreviations: K_a , Rate of non-synonymous Substitutions; K_s , Rate of synonymous Substitutions; MEGA, Molecular Evolutionary Genetics Analysis; MEME, Multiple Expectation Maximization for Motif Elicitation; MYA, Million Years Ago; QTL, Quantitative Trait Locus; TF, Transcription Factor.

* Corresponding author at: Department of Plant Breeding and Genetics, University of Agriculture Faisalabad, Faisalabad-38040, Pakistan.

E-mail addresses: rashidpbg@uaf.edu.pk (M.A.R. Rashid), tehsenpbg@uaf.edu.pk (M.T. Azhar), iqrar_rana@uaf.edu.pk (I.A. Rana), farrukh@gcuf.edu.pk (F. Azeem), chung@chonnam.ac.kr (G. Chung), zulfiqar.ali@mnsuam.edu.pk (Z. Ali), dratif@uaf.edu.pk (R.M. Atif).

¹ These authors contributed equally to this work.

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1. Introduction

The gene expression in living organisms is primarily regulated by transcription factors (TFs). There are various kinds of TF families in plants each with a role in gene expression at various developmental phases and under various stress environments. The AP2/ERF (APETALAA2 / Ethylene responsive factors) superfamily is one of the most important and largest class of plant TFs that has been involved in regulation of gene expression under diverse biotic and abiotic stresses [1,2]. AP2/ERF TF family members contain a characteristic AP2 DNA-binding domain with a length of 60–70 amino acids. Based on the number and position of AP2 domains, the AP2/ERF super-family has been further classified into five subfamilies (i.e. ERF, DREB (dehydration-responsive element binding), AP2, RAV (Related-to-ABI3/VP1), and soloist families). The ERF family members are characterized by the presence of one AP2 domain and one ERF domain. DREB family members can be identified by the presence of a single AP2 domain along with DREB domain. Similarly, RAV members contain a B3 domain along with the AP2 domain and, AP2 subfamily members are characterized by the presence of one or two AP2 DNA-binding domains [3]. Both the AP2 domains share a similarity of 25 conserved amino acids [4]. The AP2 subfamily can be further categorized into two groups i.e. euAP2 and ANT (AINTEGUMENTA) [4,5].

The AP2 subfamily members are involved in several molecular pathways associated with plant growth, development and response to various environmental stresses by regulating the target genes expression [2,6]. The modern land plants consist of variable numbers of AP2 encoding genes. These genes would have been evolved from unicellular green algae (*Chlamydomonas reinhardtii*) which exhibited only one AP2 encoding gene [7]. Likewise, the AP2 domain has also been reported to share the homology with bacterial and viral endonucleases [8]. In angiosperms, AP2 TFs have also been reported to perform diverse functions like floral meristem formation, floral organ specification, floral homeotic gene expression, ovule development and floral organs growth [6,9,10].

In *Arabidopsis*, the biosynthesis of storage compounds at the time of seed development is mediated by various AP2 genes [11]. Wrinkled 1 that has been considered as a member of a third monophyletic group of AP2 subfamily [12], has been shown as one of the most important regulators of oil accumulation in the developing seeds [13]. Moreover, AP2 TFs have also been reported to take part in various other processes like glycolytic and fatty acid biosynthesis [14,15], regulation of germination and seedling growth [16], suppression of flowering when targeted by microRNA miR172 [17], leaf epidermal cell identity [18], and embryo development [19]. Pathways of cold-responses and ovule development in Chinese cabbage are also primarily regulated by AP2/ERF family [20]. To date, AP2 subfamily members have been identified in various plant species e.g. barley [21], cotton [22], desert moss [23], sunflower [24] and pepper [25].

Rice (*Oryza sativa* L.) is a very important cereal crop serving as a staple food for >50% the world population, particularly in Asia [26]. The cultivated rice consists of two subspecies: *O. sativa* subsp. *indica* and *O. sativa* subsp. *japonica*; the former accounts for >70% of total rice production across the world. Identification and characterization of economically important AP2 genes and proteins in *indica* rice can play a vital role in ensuring food security. In the present study, systematic efforts were made to identify the genes from AP2 subfamily in *indica* rice. The evolutionary history, gene structure, conserved domains and other important information related to these genes in comparison with model plant *Arabidopsis* were also discussed in this manuscript. To further reveal its role in developmental and other regulatory processes, we surveyed the expression data and analyzed the genome-wide expression of AP2 genes from leaf tissues of rice.

2. Materials and methods

2.1. Database search, sequence retrieval and comparative analysis

The whole genome sequence of *Oryza sativa* ssp. *indica* (genome assembly ASM465v1 of cultivar 93–11) along with general feature format file (GFF3) was downloaded from the online web server (http://public.genomics.org.cn/BGI/rice_seq) and extensively searched for AP2 subfamily members using CLC Sequence Viewer version 7 (<https://digitalinsights.qiagen.com>) [27]. Moreover, Arabidopsis AP2 protein sequences were retrieved from the Arabidopsis Information Resource (TAIR) (TAIR: <https://www.arabidopsis.org/>) and BLASTed against *indica* rice genome. The protein sequences of resultant *indica* rice AP2 transcription factors were downloaded and duly verified from plant transcription factor database (<http://planttfdb.cbi.pku.edu.cn/>) [28]. The signature AP2 domain for each predicted protein was verified using NCBI web-based conserved domain search tool using conserved domain database (<https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi>) [29]. The AP2 peptide sequences were further subjected to the ExPASy server for the determination of molecular weights and isoelectric points (https://web.expasy.org/compute_pi/) [30].

The identified AP2 genes from *indica* rice were further compared with *japonica* rice AP2 genes. For this purpose, the latest genome assembly of *japonica* rice (IRGSP-1.0 of cultivar Nipponbare) along with GFF3 file was downloaded. Both *indica* and *japonica* genome assemblies were used to estimate the synteny and collinearity and draw dual synteny plot using in-built MCScanX program [31] in TB Tools (v1.045) [32].

2.2. Chromosomal location and gene structure analysis OsAP2 genes

The data about the start and the end positions of each of the identified OsAP2 gene were retrieved from the Plant Transcription factor database duly verified from the GFF3 file. These positions were also verified along the chromosomes from downloaded rice genome by using CLC sequence viewer version 7 (CLC Bio, QIAGEN A/S, Aarhus, Denmark) [27]. Finally, the OsAP2 genes were mapped on the *O. sativa* ssp. *indica* chromosomes according to scale using MapChart v2.32 [33]. Afterward, the arrangement of intron/exons of OsAP2 genes was investigated by comparing coding and genomic sequences through gene structure display server GSDS v 2.0., [34] accessible online at <http://gsds.cbi.pku.edu.cn/>.

2.3. Classification of OsAP2s and prediction of paralogous gene pairs

Amino acid sequences of all OsAP2 subfamily members were aligned through a multiple sequence alignment tool Clustal Omega (<https://www.ebi.ac.uk/Tools/msa/clustalo/>) [35] using default parameters. The resultant multiple sequence alignment was then used to infer the evolutionary history of OsAP2s by the Neighbor-Joining method [36] using Molecular Evolutionary Genetics Analysis (MEGA version 7) software suite [37]. All the positions with gaps and missing data were removed from the final data set. Poisson correction method was used to measure the evolutionary distances based on the supposition of equal substitution rates between sites as well as equivalent amino acid frequencies [38]. The reliability of resultant phylogenetic tree was assessed through 1000 bootstrap replicates. The paralogous gene pairs (PGPs) were identified from the final tree as those proteins which were present in the form of the pair on the terminal nodes, well supported by their bootstrap values.

2.4. Inference of dates of duplication of paralogous OsAP2 genes

Pairwise alignment of the coding sequences of PGPs was performed through ClustalW, using PAM protein weight matrix in MEGA software. The resultant alignment was loaded to DNA sequence polymorphism software (DnaSP v5.10.01 [39]). After defining the coding region, the synonymous substitution rate (K_s) and non-synonymous substitution rate (K_a) of each PGP were determined. The K_a/K_s ratios were then calculated for the measurement of selection pressure on amino-acid substitutions during the evolution and expansion of given sequences [40]. The value of K_s was then further exploited to estimate the approximate evolutionary dates of the segmental and tandem duplication events by using formula $T = K_s/2\lambda$, considering clock-like rates (λ) of synonymous substitution of 6.5×10^{-9} substitutions / synonymous site/year for rice [41].

2.5. Conserved domain analysis

MEME suit (Multiple EM for Motif Elicitation) available online tool (<http://meme-suite.org/tools/meme>) was used to identify the conserved domains (motifs) and related duplication events during the evolution of OsAP2 proteins, using the complete sequence of amino acids [42]. Default settings of parameters were used with following exclusions. The probability of occurrence of motifs was any number per sequences and 19 was the maximum number of motifs to be identified. The maximum and minimum width of the motif was adjusted to 12 and 50 residues respectively. The minimum number of sites per motif was adjusted to 4. The corresponding consensus sequences of conserved domains were also identified. The evolutionary relationship based on identified domains was shown in the graphical form.

Table 1

Detailed information of AP2 subfamily members in *indica* rice.

Sr. #	Gene-ID	Common Name	Chr. No.	New names	Strand	Start (bp)	End (bp)	Gene Size (bp)	CDS size (bp)	Protein Size (AA)	Mol. Wt. (KDa)	PI
1	BGIOSGA004677-PA	Osl_04180	1	OsAP2-1.1	(+)	37,996,391	37,999,162	2771	1281	426	46.357	8.88
2	BGIOSGA000316-PA	Osl_04803	1	OsAP2-1.2	(-)	42,803,611	42,807,285	3674	2085	694	71.702	6.56
3	BGIOSGA008619-PA	Osl_08060	2	OsAP2-2.1	(+)	26,018,809	26,022,976	4167	2103	700	73.213	6.31
4	BGIOSGA011350-PA	Osl_10229	3	OsAP2-3.1	(-)	4,461,575	4,465,723	4148	1641	546	58.105	7.16
5	BGIOSGA011170-PA	Osl_10636	3	OsAP2-3.2	(-)	7,457,580	7,461,302	3722	1920	639	68.287	6.29
6	BGIOSGA010867-PA	Osl_11286	3	OsAP2-3.3	(-)	12,367,743	12,372,566	4823	1290	429	45.682	8.85
7	BGIOSGA013636-PA	Osl_13681	3	OsAP2-3.4	(+)	36,069,814	36,073,178	3364	1941	646	59.989	6.59
8	BGIOSGA013826-PA	Osl_14083	3	OsAP2-3.5	(+)	38,710,166	38,713,472	3306	1305	434	47.293	7.75
9	BGIOSGA016726-PA	Osl_16549	4	OsAP2-4.1	(+)	23,840,927	23,844,803	3876	1980	659	74.954	6.21
10	BGIOSGA017250-PA	Osl_17685	4	OsAP2-4.2	(+)	32,068,242	32,070,867	2625	1383	460	48.506	6.5
11	BGIOSGA017263-PA	Osl_17717	4	OsAP2-4.3	(+)	32,401,804	32,405,677	3873	1464	487	52.558	5.83
12	BGIOSGA018911-PA	Osl_18248	5	OsAP2-5.1	(-)	1,223,237	1,226,640	3403	1554	517	56.030	6.6
13	BGIOSGA019793-PA	Osl_19817	5	OsAP2-5.2	(-)	19,987,735	19,992,697	4962	1047	348	39.048	8.75
14	BGIOSGA017664-PA	Osl_20775	5	OsAP2-5.3	(-)	28,007,910	28,009,771	1861	1290	429	47.450	8.47
15	BGIOSGA021927-PA	Osl_21635	6	OsAP2-6.1	(-)	2,723,473	2,726,350	2877	1194	397	42.978	7.67
16	BGIOSGA020813-PA	Osl_23840	6	OsAP2-6.2	(-)	27,178,841	27,182,033	3192	1317	438	47.527	6.84
17	BGIOSGA020812-PA	Osl_23841	6	OsAP2-6.3	(-)	27,189,491	27,192,298	2807	1212	403	45.336	6.22
18	BGIOSGA023402-PA	Osl_23971	6	OsAP2-6.4	(+)	28,687,948	28,689,551	1603	657	218	23.618	6.28
19	BGIOSGA025077-PA	Osl_24741	7	OsAP2-7.1	(+)	1,388,211	1,394,201	5990	1677	558	59.214	7.23
20	BGIOSGA025426-PA	Osl_25450	7	OsAP2-7.2	(+)	7,599,098	7,602,786	3688	1311	436	47.433	6.8
21	BGIOSGA027579-PA	Osl_27983	8	OsAP2-8.1	(-)	4,423,677	4,426,023	2346	822	273	29.541	6.27
22	BGIOSGA028751-PA	Osl_29392	8	OsAP2-8.2	(+)	22,907,012	22,910,811	3799	1266	421	45.746	5.41
23	BGIOSGA030783-PA	Osl_31417	9	OsAP2-9.1	(+)	13,859,899	13,864,122	4223	1140	379	41.024	8.62
24	BGIOSGA034580-PA	Osl_34947	11	OsAP2-11.1	(-)	1,066,992	1,071,970	4978	1332	443	49.218	4.78
25	BGIOSGA035153-PA	Osl_35861	11	OsAP2-11.2	(+)	9,547,766	9,552,408	4642	1551	516	55.757	6.31
26	BGIOSGA036704-PA	Osl_37329	12	OsAP2-12.1	(-)	1,613,154	1,618,116	4962	1332	443	49.305	4.82

2.6. Prediction of miRNAs target

Revealing of micro RNA (miRNA) target genes have a significant role to understand the functions of miRNAs. *In-silico* prediction of miRNAs potentially targeting rice AP2 genes was carried out by submitting the transcript sequences of OsAP2s on the web-based psRNA Target Server (<https://plantgrn.noble.org/psRNA/Target/>). Stringent selection criteria with a penalty score of ≤ 2.0 were adopted for the identification of targets using the already reported rice miRNA database available at the above-mentioned web server. This miRNA database contains detailed information on 713 published miRNAs of rice. The stringent selection criteria are helpful in the identification of potential targets of miRNAs having high specificity and to avoid the background noise in target prediction.

2.7. Insilico expression analysis of annotated OsAP2 genes

To investigate the expression pattern of AP2 genes in rice, the transcriptomic data was collected from the Rice Genome Annotation Project (Project) at different developmental stages from different tissues. Later, the expression profile was drawn from transcriptomic data. Finally, a heat map was generated by XLSTAT software.

2.8. Plant materials and stress treatment

A rice heat tolerant mutant “HTT-112” and a heat sensitive mutant “HTT-1” reported previously by one of our authors was used for gene expression profiling in this study [43]. Both the heat tolerant and heat sensitive mutants were grown under control conditions at $28 \pm 2^\circ\text{C}$ for 3 weeks. The mutants were subjected to 24 h of heat stress at $45 \pm 2^\circ\text{C}$ afterwards. The leaves were harvested from control and treated samples and stored at -80°C before RNA isolation.

2.9. RNA isolation and quantitative real time PCR

Total RNA was isolated from collected leaves using RNA-prep pure plant kit (TIANGEN) and reverse transcribed using HiScript II QRT SuperMix (Vazyme) [44]. Quantitative real-time PCR for 26 AP2 genes was performed using SYBR green mixture on an ABI 7500 real-time PCR detection system. Rice Ubiquitin gene was used as internal control for normalization [44]. All primer sequences are listed in Supplementary table (Table S1).

2.10. Comparative phylogenetic analysis

For comparative analysis, the peptide sequences-based multiple sequence alignment of 28 AP2 proteins from *Brassica rapa*, 18 AP2s from *Arabidopsis thaliana*, 26 AP2s from *Oryza sativa*, and 1 from *Chlamydomonas* was carried out with the help of Clustal Omega (<https://www.ebi.ac.uk/Tools/msa/clustalo/>) [45]. Moreover, the only AP2 gene from a unicellular green alga (*Chlamydomonas reinhardtii*), which is believed to be the ancestor of this family in modern land plants was also included in the comparative analysis to root the tree. The Neighbor-Joining method was used to infer the evolutionary history [36]. The Poisson correction procedure was used to measure the evolutionary distances [38] in units of the number of amino acid substitutions per site. Total 73 amino acid sequences were used for that analysis. All positions having less than 95% site coverage were removed. That is, less than 5% alignment gaps, missing data, and ambiguous bases were allowed at any position. Finally, the dataset was composed of total 140 positions. MEGA7 was used to perform evolutionary analysis [46].

2.11. Haplotype analysis

The mini-core collection of 305 cultivated rice genotypes possessing the 70.65% and 76.97% SSR and phenotypic variation of 4310

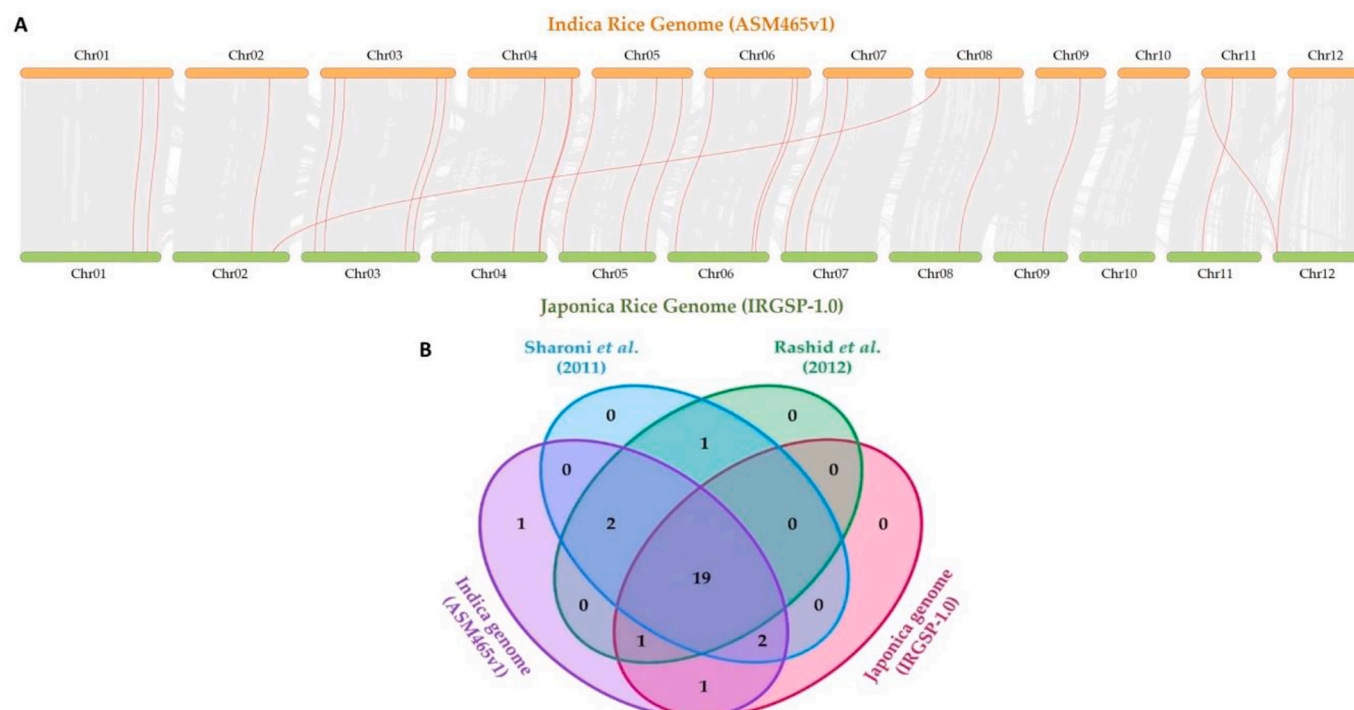


Fig. 1. Comparison of OsAP2 genes between *indica* and *japonica* rice genomes. A. Genome-scale dual synteny plot between *indica* and *japonica* rice genomes, with orthologous OsAP2 genes shown in red lines. B. Venn diagram showing comparison of OsAP2 genes reported in this study with previous studies. Sharoni et al. and Rashid et al. reported 24 and 23 AP2 genes respectively in *japonica* rice. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

accessions respectively was used for haplotype analysis [47]. The whole genome sequences were obtained and *indica* genotypes were selected based on population structure for haplotype study [48]. The total length of transcriptional factors was evaluated for SNPs extraction. The accessions having missing SNPs >10% were eliminated and finally, the minimum 141 genotypes for (LOC_Os05g03040.1) to maximum 165 genotypes for (LOC_Os01g67410.1) were evaluated for haplotype analysis. DnaSP was used for the evaluation of haplotypes and haplotype diversity (Hd) [39].

3. Results and discussion

3.1. Identification of AP2 subfamily members in *Oryza sativa* ssp. *indica*

A systematic approach was followed to identify the AP2 TF subfamily-encoding genes using the publicly available genome datasets. For this purpose, the sequenced genome of *O. sativa* ssp. *indica* was downloaded and scanned for the identification of genes encoding AP2 TF subfamily. Initially, 28 AP2 genes were identified and their sequences were retrieved from online available databases as well as from downloaded rice genome. Further, verification of these sequences had led us to exclude two sequences because of their truncated AP2 domains. Finally, 26 non-redundant genes encoding complete AP2 DNA-binding domains were selected for further analysis. Nomenclature of these genes was carried out according to the position of the genes on the chromosomes as suggested in other plants [49,50]. Renaming of these genes was done as *OsAP2* (*Oryza sativa* AP2) with chromosome number followed by decimals for the multiple genes on respective chromosome. For example, *OsAP2-1.1* refers to first AP2 gene present on chromosome 1, while *OsAP2-3.5* refers to 5th AP2 gene present on chromosome no. 3 (Table 1). The detailed information of these genes including their TF-ID, CDS length, protein size and molecular weight, chromosome number, location, and size of the full-length gene are given in Table 1. The peptide length of deduced proteins varied from 273 amino acids (AA) (*OsAP2-8.1*) to 700 AA (*OsAP2-2.1*), with corresponding molecular weights ranged from 29.54 to 73.21 kDa with an average weight of 51.98 kDa. The predicted isoelectric points of the identified proteins varied from 4.78 (*OsAP2-11.1*) to 8.85 (*OsAP2-3.3*).

3.2. Comparison of copy number of *indica* AP2s with *japonica* and other plants

The genome of *O. sativa* subsp. *japonica* was also thoroughly searched

for the identification of AP2 subfamily members. Initially 25 genes were identified, 2 among them (*Os03t0770700* and *Os11t0129700*) lacked complete AP2 domains, so they were excluded from the list. Finally, we report 23 genes encoding AP2 subfamily members in *japonica* genome (Table S2). Dual synteny plot-based in-depth comparative analysis of *OsAP2s* among these *indica* and *japonica* genomes revealed that all the 23 *japonica* AP2 member have their homologs in *indica* genome as well (Fig. 1A, Fig. S1). On contrary, homologs of 3 *indica* AP2s (*OsAP2-3.4*, *OsAP2-6.2* and *OsAP2-11.1*) could not be identified in *japonica* genome. Two *japonica* genes that were excluded after initial screening i.e. *Os03t0770700* and *Os11t0129700* shared some homology with *OsAP2-3.4* and *OsAP2-11.1* respectively but lack the complete domains. These two genes might become pseudogenes either due to frame shift mutations or pre-mature stop codons [51].

Previously, 24 genes belonging to the *japonica* AP2 subfamily have been reported by Sharoni and colleagues in 2011 [52], while 23 *japonica* AP2 subfamily members were identified by Rashid and colleagues in 2012 [53]. Further comparative analysis of AP2s among these two reports along with the latest *indica* and *japonica* genome assemblies enables us to identify 19 common genes (Fig. 1B; Table S1). The report of 2011 shared one gene (*LOC_Os08g07440*) with report of 2012, which could not be found in latest assemblies of both *indica* and *japonica* genomes. Although *LOC_Os08g07440* shares homology with *Os08g0171100* in *japonica* assembly IRGSP-1.0 but *Os08g0171100* looks like a pseudogene because it encodes a small truncated protein (60 AA peptide length) with only a small portion of AP2 domain. This might be the result of mis-annotation of “pseudogenes” as “functional genes” in the earlier rice genome assemblies [54]. It has already been reported that some pseudogenes can have a detectable open reading frame despite having premature stop codon and are thus distinct from those pseudogenes that typically lack defined open reading frames and are identified in the intergenic regions [54]. These pseudogenes can be resulted either due to retrotransposons or due to decay of genes originated from duplication event [55].

Two AP2 genes identified by Sharoni and colleagues [52] i.e. *LOC_Os02g40070* and *LOC_Os04g42570* were not reported by Rashid et al. [53] because both carry only one AP2 domain and hence excluded from AP2 subfamily by Rashid et al. One gene in our study i.e. *OsAP2-6.2* (BGIOGA020813) is unique and its homolog could not be found in either of remaining three cases. Its detailed analysis demonstrated that this gene would have resulted from tandem duplication of another gene *OsAP2-6.3* (BGIOGA020812) as both are present on chromosome 06 next to each other.

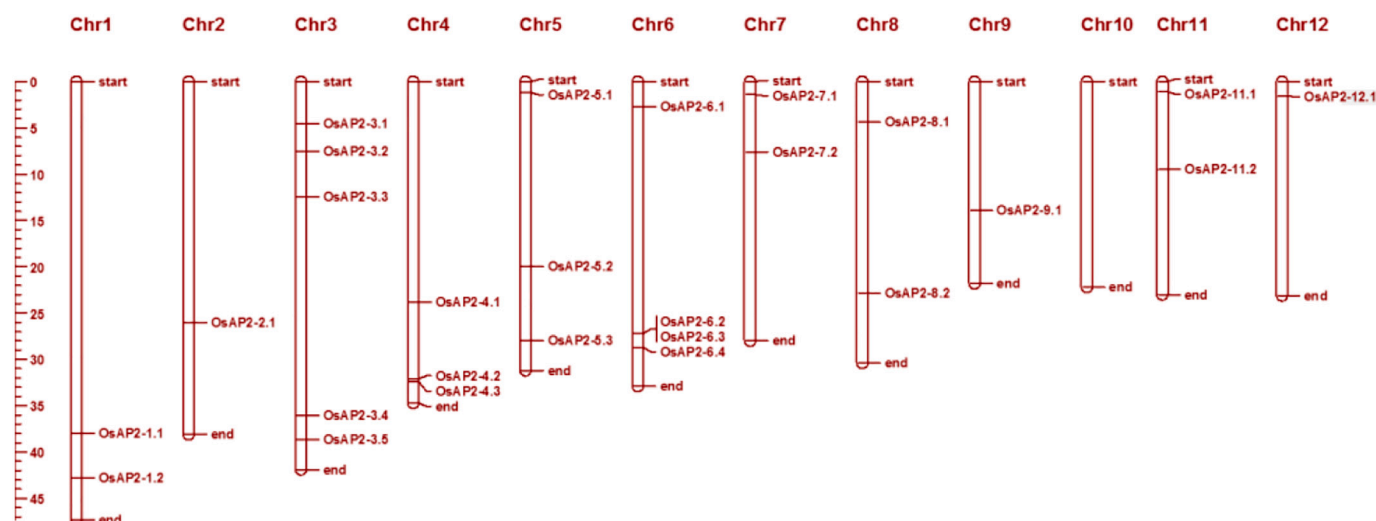


Fig. 2. The localization of 26 *OsAP2* genes on *indica* rice chromosomes. The chromosome numbers are indicated at the top of each bar. Gene names correspond to those listed in Table 1. The relative position of *OsAP2s* and size of chromosome represented using a vertical scale in Mb.

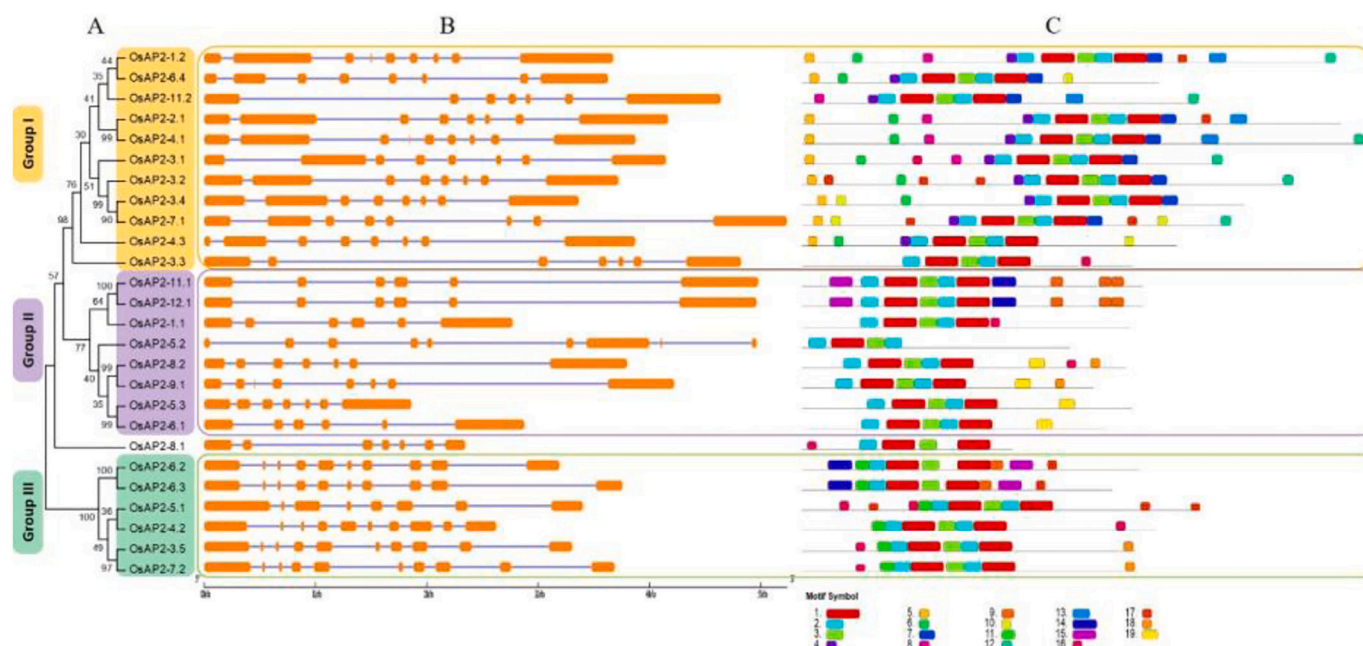


Fig. 3. Phylogenetic tree, Gene Structure and conserved domain analysis of *OsAP2* genes in *Oryza sativa* ssp. *indica*. (A) The tree is divided into three groups indicated as Group I–III. The percentage of replicate trees in which the associated taxa clustered together in the bootstrap test (1000 replicates) is shown next to the branches. All the AP2 protein sequences are used for the conserved domain and duplication analysis using online MEME server. (B) Intron/exon arrangement-based gene structure of *OsAP2* genes. (C) 19 conserved motifs of different lengths are depicted on the protein map next to the phylogenetic tree. The motif codes are indicated below the figure and the peptide sequences of the respective domains are shown in Table S3.

Several other studies reported a varying number of AP2 family members in different plants. For example, 18 genes belonging to AP2 subfamily were reported in *Arabidopsis* [10], whereas 20 in grapevine [56], 26 in poplar [57], 21 in Peach [58], 19 in castor [6], 35 in sunflower [24] and 29 in pepper [25]. Similarly, 32 and 30 genes were assigned to the AP2 subfamily in *Brassica oleracea* and *B. rapa* respectively [59]. This diversity in gene copy number variation of AP2 subfamily has been proposed as a potential driver of morphological diversity in angiosperms [60].

3.3. Chromosome localization of *OsAP2s*

The distribution of predicted *OsAP2s* was virtually localized on *O. sativa* chromosomes. The distribution of predicted 26 *OsAP2* genes was non-random across 11 out of 12 *O. sativa* chromosomes (Fig. 2). This non-random sort of gene distribution has also been reported in other eukaryotes [61]. Long terminal repeat retrotransposons have been proposed to be responsible for the evolution and nonrandom distribution of rice gene families across the rice genome [62,63].

The maximum 5 and 4 AP2 genes were located on single chromosome 3 and 6 respectively. Chromosome 2, 9 and 12 carried only one AP2 gene each while none of the AP2 genes was present on chromosome 10. The smallest chromosome was 9 with only one gene while chromosome 1 is recognized as largest among all chromosomes and carried two genes. Few genes were present far from each other on the same chromosome, while some were found in the cluster form and revealed that genes with similar groups may carry common and/or single QTL. When compared with the distribution of AP2 genes in *japonica* genome, all the 23 AP2 homologous gene pairs shared the conservation of localization in both *indica* and genomes except the pair *Os02t0747600* and *OsAP2-8.1*. Thus, two AP2 genes of *japonica* were present on chromosome 02 in *japonica* genome (*Os02t0614300* and *Os02t0747600*), while only one AP2 was identified in *indica* genome i.e. *Ap2-2.1* which is homolog of *Os02t0614300*. It is interesting to note that homolog of *Os02t0747600* (present on chromosome 02) in *indica* was identified as *OsAP2-8.1* which was present on chromosome 08. This presence of two

homologous genes of two subspecies on different chromosomes might have resulted from structural variation caused by inter-chromosomal translocation, as reported recently in *indica* and *japonica* rice genotypes [64,65].

Previously, various studies of AP2 subfamily members had exhibited the uneven distribution of these genes across the genomes of plant species. Some chromosomes of these plants were preferred for the extension of this family. For example, in *Populus trichocarpa*, 25 out of 26 *PtAP2* genes were dispersed unequally on 12 out of 19 chromosomes, while one *PtAP2* was located on un-assigned scaffold region [66]. Likewise, in the case of *Vitis vinifera*, 18 genes of AP2 subfamily were distributed on 10 out of 19 chromosomes with three genes only on chromosome number 6, two genes each on chromosome number 4, 7, 8, 9, 11 and 18, and one AP2 gene each on chromosome 1, 13 and 14 [56].

Brassica rapa had a total of 29 AP2 genes located on 10 chromosomes [20]. Chromosome 9 was the largest while chromosome 10 was the smallest among all chromosomes of *B. rapa* with two AP2 genes on each. In *B. oleracea*, there was a total of 32 AP2 genes located on 9 chromosomes [5]. The maximum five AP2 genes were present on chromosome 2. Chromosome number 1, 4, 5, 6 and 8 had two AP2 genes each and chromosome 3 and 9 had three AP2 genes on each of them. Four AP2 genes were located on chromosome 7, 7 AP2 genes were present in scaffold regions. Chromosome 3 was the largest; in contrast, chromosome 5 was the smallest among all chromosomes of *B. rapa* with three and two AP2 genes located on each chromosome respectively.

3.4. Phylogenetic analysis of *OsAP2* transcription factor family

An unrooted phylogenetic tree was built to explore the evolutionary relationship of the AP2 family members in *indica* rice. The phylogenetic tree exhibited the distribution of *OsAP2* transcription factors in three distinct groups (Fig. 3A). The distribution of AP2 subfamily members into three groups is due to conservation of group-specific domains among the proteins.

Group I represents the largest group with 11 out of 26 *OsAP2* proteins but their physiological roles have not functionally characterized yet. The

Table 2

Duplicated AP2 genes along with their estimated dates of duplications in rice genome.

Sr. #	Locus 1	Chr #	Locus 2	Chr #	Duplication type	Ka	Ks	Ka/Ks	T = Ks/2λ
1	OsAP2-1.2	1	OsAP2-6.4	6	Segmental	0.30	1.98	0.15	22.98
2	OsAP2-2.1	2	OsAP2-4.1	4	Segmental	0.18	0.46	0.39	13.91
3	OsAP2-3.4	3	OsAP2-7.1	7	Segmental	0.27	0.57	0.47	20.76
4	OsAP2-3.5	3	OsAP2-7.2	7	Segmental	0.17	0.90	0.19	13.40
5	OsAP2-4.2	4	OsAP2-5.1	5	Segmental	0.43	1.22	0.36	33.41
6	OsAP2-5.3	5	OsAP2-6.1	6	Segmental	0.41	2.69	0.15	31.40
7	OsAP2-8.2	8	OsAP2-9.1	9	Segmental	0.27	0.73	0.37	21.06
8	OsAP2-11.1	11	OsAP2-12.1	12	Segmental	0.74	3.26	0.23	56.72
9	OsAP2-6.2	6	OsAP2-6.3	6	Tandem	0.14	0.19	0.74	10.57

Hence, K_a and K_s indicate the non-synonymous and synonymous substitutions respectively; T represents the segmental and tandem duplication events, while λ was considered as clock-like rates of synonymous substitution as 6.5×10^{-9} substitutions / synonymous site/year for rice.

members of group I shared some unique motifs among multiple sequence alignment e.g. NSCRREGQTRK, XTITTTNFPXNN and TRHHQHGRWQ (Fig. S2). Eight OsAP2 proteins were present in Group II with variable forms of motifs e.g. KNXXXXXXXXKK, PETVLNFPLEE and ARHHHNGRWE (Fig. S2). Group III had only six OsAP2 proteins with lacking the domain 1 and variable motifs i.e. LEADINFNLSD and TLH-KCGRWE (Fig. S2). This classification of AP2 subfamily into three distinct groups is absolutely conserved in all the monocots and dicots whose genome has been sequenced to date. This conservation revealed the evolution of AP2 subfamily groups before the divergence of gymnosperm and flowering plant ancestries [7]. OsAP2-8.1 is the one distinct member of AP2 family which is evolutionarily distinct from rest of family members and do not fall into three identified groups (Fig. 3A). The multiple alignment analysis of amino acids infer that it may due to the specific miss-sense and insertion/deletion mutations, which cause the transition, addition and deletion of amino-acids in conserved regions (Fig. S2).

The distribution pattern of the OsAP2 protein in three clades in the phylogenetic tree is in accordance with the already reported trees, e.g. phylogenetic tree of 19 AP2 members of castor bean divided them into 3 groups and one member shared none of the groups [6]. In common bean, 27 genes of the AP2/ERF family having double AP2 domains were classified into the AP2 subfamily. The phylogenetic analysis then divided the AP2 subfamily into three clades or subgroups [50]. In *musa* bamboo, 28 genes of the AP2 subfamily were assembled into three clades, i.e. 7 genes in the first clade, 6 genes in the second, 13 genes in the third clade and 2 genes were not included in any of identified clades [67]. A total of 27 rice AP2 genes were distributed into three clades named as Ia, Ib and Ic, having 10, 11 and 6 genes respectively [68]. In parsley, AP2 subfamily contains 12 gene members which were divided into three clades having three genes in clade 1 and 2 each while six genes in clade 3 [69].

3.5. Gene structure analysis

The coding and genomic DNA sequences of every one of OsAP2 genes were compared for the determination of numbers and positions of exons and introns. Gene structure analysis showed that OsAP2 genes have multiple introns ranging from 4 to 9 with predominantly 7 introns in the 9 OsAP2 genes (Fig. 3B). On contrary, AP2 subfamily members in *Japonica* rice exhibited five to 114 introns [52]. Our results further revealed that only one gene contains 4 introns, four genes contain 5 introns, five genes contain 6 introns, 3 genes contain 8 introns and 5 genes contain 9 introns. Similar gene structure pattern has been reported in AP2 subfamily members in castor bean, which contained 3 to 9 introns [6]. The gene structure comparison based on phylogenetic tree revealed the presence of 9 introns in all the members of group 3 except OsAP2-5.1 which possessed 7 introns.

Moreover, comparison of gene structure between *indica* and *japonica* showed the overall conservation of intron/exon arrangements among AP2 genes of both species with few exceptions. For example, OsAP2-2.1 has 7

introns in *indica* while its homolog Os02t0614300 has only 3 introns (Table S2). Thus, in the case of OsAP2 family members, some intron loss events might have occurred during evolution. It has been reported that during the evolution of eukaryotic lineages, intron losses have outnumbered the intron gains [70]. Interestingly, it was also noticed that all the PGPs exhibited comparable gene structure, mostly sharing the same number and arrangement of introns/exons.

In *musa* bamboo, 1–9 introns were present in AP2 subfamily members, demonstrating the possible intron losses during evolution in some of the AP2 family members [67]. On the other hand, the gene structure of Arabidopsis AP2 subfamily revealed the presence of 6 to 9 introns with predominantly 6 introns in seven AtAP2 genes [10]. The number of introns among AP2 subfamily members varied from 5 to 9 and approximately half of the group members have seven introns. PvAP2-ERF179 is the only member of AP2 subfamily which had five introns [50].

3.6. Identification of conserved domains in OsAP2 proteins

Conserved domain analysis was performed through MEME server by using the whole protein sequences of OsAP2 members. As a result, there were a total of 19 diverse conserved domains has been recognized among 26 AP2 members (Fig. 3C). The range of amino acid length among identified domains was varied from 12 to 43 (Table S3). As shown in the phylogenetic tree, few groups contain 3–4 conserved domains while other groups with multiple domains. The number of conserved domains was different among three groups as, 6–7 (Motif 1, 2, 3, 4, 5, 6, and 11), 3 (Motif 1, 2 and 3) and 5 (Motif 1, 2, 3, 10 and 17) in group I, II, III respectively. Group I and III had highly conserved domains i.e. Motif 1, 2, 3. The presence of multiple conserved domains in OsAP2s proteins proposes that the genes encoding these proteins create from the similar duplication events. Three highly conserved domains (Motif 1, 2 and 3) have been found among all the OsAP2 proteins members.

These genes encoding for that proteins also had the same position on the same chromosomes. In *O. sativa*, all the 26 AP2 proteins share the 43 residues conserved domain (motif 1, NKLYLGTFTDEEEAAR-AYDLAAIKYRGLNAVTFNFDLSYEKE) as well as 22 residues conserved domain (motif 2, SIYRGVTRHRWTGRWEAHIWDV) and the 22 residues conserved domain (motif 3, MKNMTREEYVASLRKSSGFSR). The presence of conserved domains among different AP2 proteins especially in the same group likely reflects related functions of these proteins. The proteins carried the same conserved domains reflects their functions similarity especially, within the groups.

The domains like motif 1 and motif 2 have two sites present in 24 proteins out of the total 26 AP2 proteins. In castor bean, two motifs 9 and 11 of the AP2 subfamily were highly conserved. The motif 9 was shared by 10 members and motif 11 was shared by 15 members of the AP2 subfamily [6]. The results of a previous study [68] showed that 23 out of 26 AP2 genes had double conserved domains or motifs while four had only one conserved motif. In *Salix arbutifolia*, AP2 subfamily had a conserved domain motif 5 in 22 genes and motif 2 in 23 genes out of

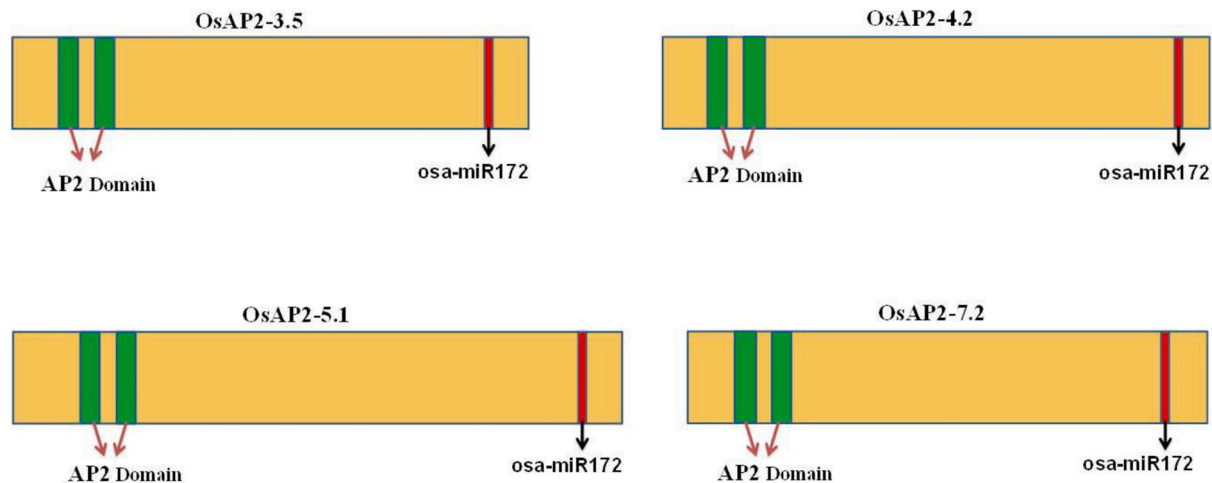


Fig. 4. Target sites of *osa-miR172* in *OsAP2* subfamily. Structure of the genes targeted by *osa-miR172* family members showing AP2 domain sites and location of the target of *osa-miR172* family members.

total 24 genes [71].

3.7. Identification of *OsAP2* paralogous gene pairs

The expansion and evolution of any gene family are usually linked with whole genome duplications, segmental (direct and/or inverted interspersed) and/or tandem duplications [72]. Segmental duplication of the chromosomal fragment has been reported to be the major contributor of gene family expansion and evolution, resulting in a dispersed occurrence of gene family members on different chromosomes [73]. Gene families can also evolve through tandem amplification of chromosomal region, resulting in clusters of duplicated genes on the same chromosomes [74].

The nature of the expansion of the AP2 subfamily in rice was analyzed by predicting the paralogous *OsAP2* gene pairs. Based on the phylogenetic tree, a total of 9 PGPs could be identified i.e. *OsAP2-1.2/OsAP2-6.4*, *OsAP2-2.1/OsAP2-4.1*, *OsAP2-3.4/OsAP2-7.1*, *OsAP2-3.5/OsAP2-7.2*, *OsAP2-4.2/OsAP2-5.1*, *OsAP2-5.3/OsAP2-6.1*, *OsAP2-8.2/OsAP2-9.1*, *OsAP2-11.1/OsAP2-12.1* and *OsAP2-6.2/OsAP2-6.3* (Table 2). All the three groups of phylogenetic tree contained three paralogous genes pairs each. Members of PGPs were either co-located on the same chromosome e.g. *OsAP2-6.2/OsAP2-6.3* both located on chromosome no. 6; or were present on different chromosomes e.g. *OsAP2-1.2/OsAP2-6.4* located on chromosome no. 1 and 6 respectively.

These PGPs present on different chromosome might have resulted from the segmental duplications because the genes sharing the synteny were found on two different chromosomes in the same species [75]. In this study, a total of 8 out of 9 *OsAP2* PGPs had resulted from the segmental duplications. Conversely, the PGPs sharing the same chromosome might have developed as a consequence of a tandem duplication occurrence because the genes sharing the synteny were found on the same chromosome [75]. Only 1 out of 9 *OsAP2* PGPs seemed to be evolved from tandem duplication. These results proposed that the segmental duplications might have primarily contributed towards the extension and evolution of the AP2 subfamily in *indica* rice.

3.8. Estimation of divergence time of rice PGPs

The divergence time of *OsAP2* PGPs was assessed by recording synonymous (K_s) and non-synonymous substitutions (K_a) rate per site per year. The value of K_a/K_s ranges from 0.15 to 0.74, as all the values were less than 1 ($K_a/K_s < 1$) (Table 2), it showed that synonymous changes are more than non-synonymous changes in the gene sequence of PGPs. This suggests that purifying selection was dominant and the main

route of the evolution of all *OsAP2* paralogous pairs. The approximate time of this duplication was inferred using the formula $T = K_s/2\lambda$. The results indicated that the duplications of the paralogous genes in rice occurred from 10.57 to 56.72 MYA. One tandem duplication of the AP2 gene pair *OsAP2-6.2* and *OsAP2-6.3* in rice originated very recently i.e. about 10.57 MYA (Million Years Ago) (Table 2). This duplication date is not too far from the divergence of *japonica* and *indica* rice. It has been reported that *indica* and *japonica* genomes diverged from one another at least 2.0 MYA [62]. The comparative analysis of *indica* versus *japonica* AP2 genes revealed that homolog of *indica* *OsAP2-6.2* in *japonica* genome might have been lost after the subspecies divergence. Some paralogous genes originated about 56.72 MYA with a mean of 24.91 MYA. On the other hand, the tandem duplication originated nearly 10.57 MYA. The split of AP2 subfamily from the lineage of ERF superfamily occurred about 418 MYA [76].

The duplication in AP2 family members in rice exhibited the similar pattern as reported in other plants. For example, in foxtail millet, the duplication of AP2 genes have resulted about 22–24 MYA and K_a/K_s values ranging from 0.03–0.15 predicted the strong purifying selection pressure [77]. The K_s value of 0.2 in mosa bamboo suggested that a large-scale duplication event of AP2 genes underwent nearly 15 MYA and K_a/K_s ratio of 0.3–0.5 predicted the strong purifying selection pressure [67]. The K_s value of 0.2–0.3 indicated that duplication of orthologous gene pairs among rice and mosa bamboo occurred about 15–23 MYA. In *P. vulgaris*, 57% of AP2 genes showed the segmental duplication events while 8% resulted from tandem duplications indicating that the segmental duplication had much more contribution than tandem duplications in the process of AP2 super family expansion [50].

3.9. Identification of miRNAs targeting AP2 members

MicroRNAs (miRNAs) are the non-protein coding sequences of RNA molecule which control the expression of genes at post-transcriptional level [78]. The miRNAs perform useful functions in plant developmental stages and responses to different stress environments. Various miRNAs have been reported post-transcriptionally targeting the AP2 transcriptional factor family members in various crops [50]. In this regard, we employed psRNA Target web server for the identification of rice miRNAs potentially targeting *OsAP2* transcripts. This web server uses an efficient computing pipeline to identify reverse complementary similarity among miRNAs and their mark transcripts. Moreover, it also distinguishes translational and post-transcriptional inhibition of the transcript's translation by miRNAs.

Our results demonstrated a total of 13 targets of *OsAP2*s carrying the

putative binding sites of miRNAs. The results further indicated that four members of osa-miR172 family i.e. osa-miR172a, osa-miR172b, osa-miR172c and osa-miR172d-3p are potentially targeting four different AP2 genes i.e. OsAP2-3.5, OsAP2-4.2, OsAP2-5.1 and OsAP2-7.2 (Table S4). These four members of the osa-miR172 family were present on chromosome number 9, 1, 7 and 2 respectively (Table S4). The AP2 genes being targeted are present in Group III of the phylogenetic tree. The targeted TF genes are distributed on four different chromosomes (3, 4, 5 and 7) of the rice genome. In-depth analysis of the target genes revealed the presence of osa-miR172 target sites in the last exons of all the four genes (Fig. 4).

The results further demonstrate that each of three miRNAs i.e. osa-miR172a, osa-miR172c and osa-miR172d-3p were possibly targeting three members of the AP2 TF family viz. OsAP2-3.5, OsAP2-5.1 and OsAP2-7.2, while only osa-miR172b was targeting all four AP2 TFs viz. OsAP2-3.5, OsAP2-4.2, OsAP2-5.1 and OsAP2-7.2. Interestingly, a recent rice degradome analysis showed that the transcripts of OsAP2-3.5 (LOC_Os03g60430.1), OsAP2-4.2 (LOC_Os04g55560.2), OsAP2-5.1 (LOC_Os05g03040.1) and OsAP2-7.2 (LOC_Os07g13170.1) were marked by the osa-miR172a [79]. This osa-miR172 mediated post-transcriptional gene expression regulation was responsible for the control of leaf senescence in rice [79]. Moreover, degradome analysis in *Brassica campestris* [80] also revealed the regulation of AP2 transcripts through miR172 family members, especially during pollen development. Thus, miR172 mediated post-transcriptional regulation of the AP2 genes seems to be conserved in angiosperms. The target sites of the miR172 are also well conserved in the AP2 family of gymnosperms [7].

The binding sites of the miR172 are present among the 2nd AP2 domain and the 3rd terminus [7]. The miR172 was also responsible for flowering characteristics including early flowering and differentiation of floral parts in *Arabidopsis* [81]. It also had a role as a transcriptional repressor in cell fate specification of AP2 in *Arabidopsis* [82]. The

miR172 has also been demonstrated to regulate the activities of several AP2 members during the organ development in *Arabidopsis* [82–84]. Besides *Arabidopsis*, AP2 family members from moss, rice, maize, sorghum and wheat genome also exhibited the conserved target binding sites of miR172a [76]. Recently, AP2 gene transcripts being targeted by miR172 were also identified in foxtail millet [85]. In soybean, miR172 has been shown to promote the root nodules [86], while miR172c has been implicated in the mediation of genes expression which is the regulator of symbiosis in common beans [87].

3.10. Genome-wide in-silico expression analysis of OsAP2 genes in different tissues

RNA-Seq Atlas of *Oryza sativa* L. offers the high-resolution gene expression data in a diverse set of 11 different tissue samples. For one of 26 genes OsAP2-6.4, the tissue level expression has not been reported. It may due to its variable spatio-temporal expression pattern. It was proposed that some SNPs in cis-regulatory elements in promoter region of this gene lead to silencing of gene expression [88,89]. RNA-seq normalized expression data were retrieved from the Rice Genome Annotation Project. Data were analyzed, and a heat map was made according to the diverse OsAP2 genes based on FPKM (Fragments per Kilobase of Transcript per Million mapped reads) values for each gene in different tissues. Five different OsAP2 genes (OsAP2-1.2, 2.1, 5.3, 6.1 and 11.2) were eliminated from analysis due to the interquartile range < 0.25). Expression analyses demonstrated that many OsAP2 genes have different tissue-specific expression. Furthermore, two different clusters of genes exhibited a high level of expression in embryonic and shoot development (Fig. 5).

Overall, OsAP2 genes have weak expression profile in leaves, seedling and anthers as compared to the other plant tissues. Mainly, some genes (OsAP2-8.1, 3.3, 12.1 and 9.1) have a high pattern of expression

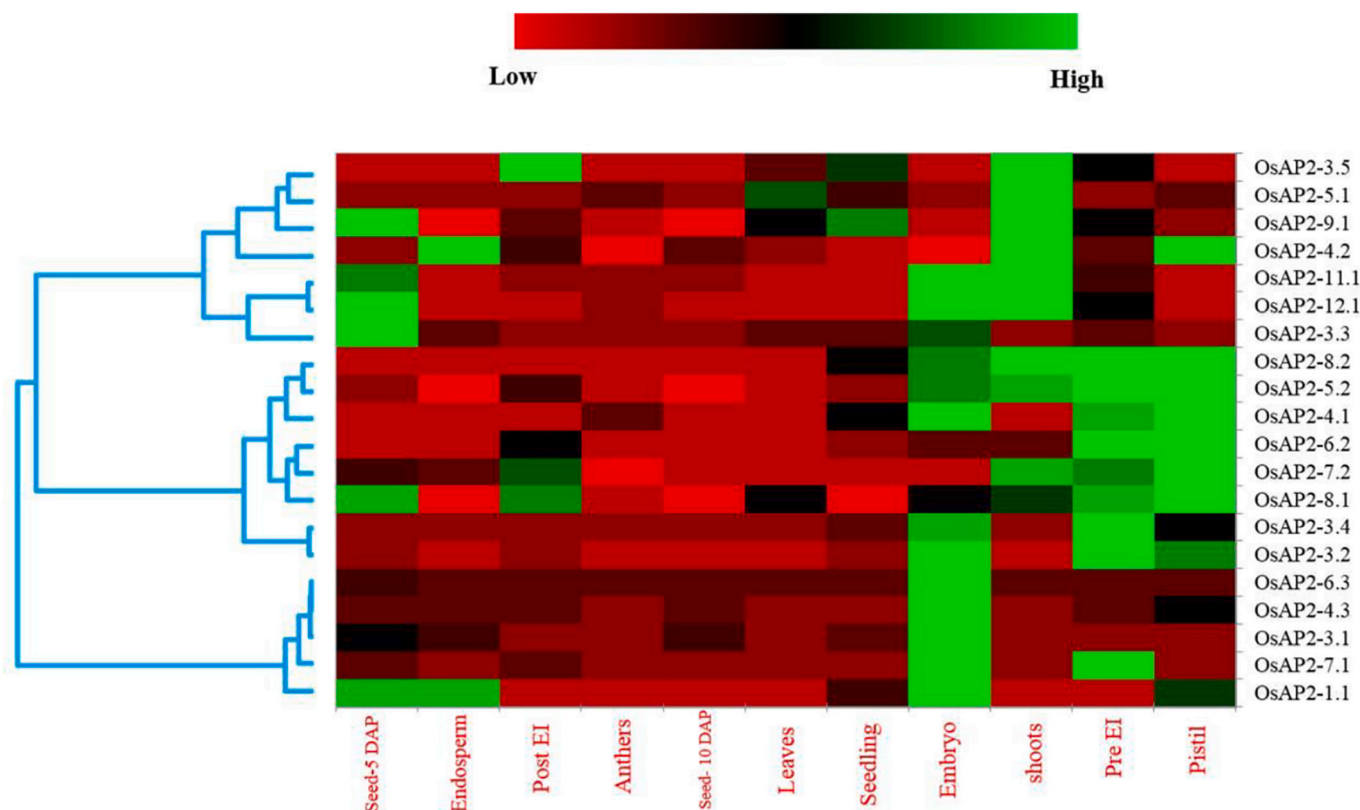


Fig. 5. Expression pattern of OsAP2 genes in different tissues at various developmental stages based on FPKM (Fragments per Kilobase of Transcript per Million mapped reads) values. The colour scheme is representing the expression intensity (red, low expression; green, high expression) in different tissues. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

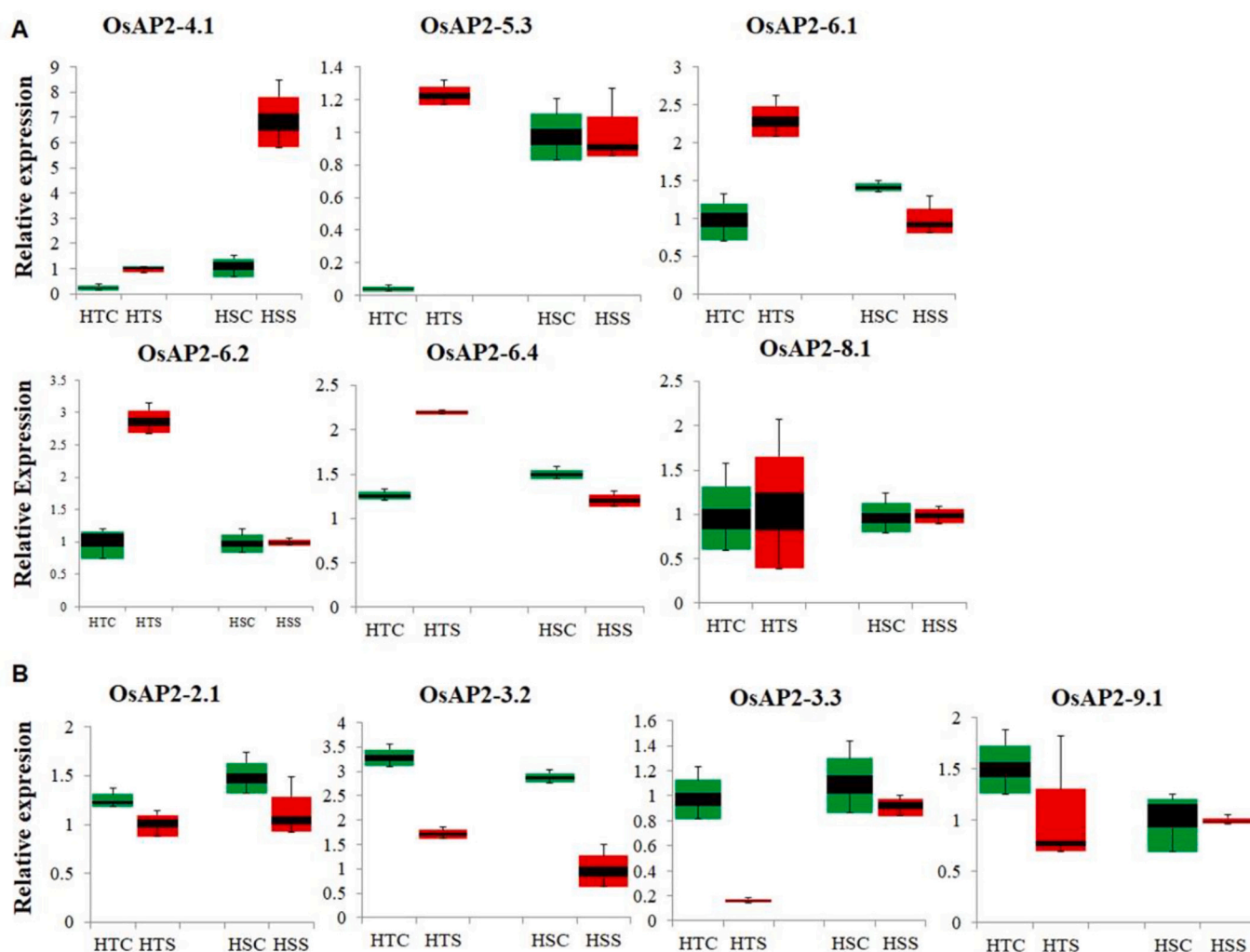


Fig. 6. Quantitative real-time PCR-based expression profiles of OsAP2 genes in response to heat shock treatment on heat tolerance genotype before stress (HTC), heat tolerance genotype after heat stress (HTS), heat susceptible genotype before stress (HSC) and heat susceptible genotype after heat stress (HSS), indicating the (A) up-regulated, (B) down-regulated genes, Error bars indicate the Mean $\pm$ SD of three replicates.

in endosperm and seed development which reveals their putative function in seed development (Fig. 5). Some genes (OsAP2-6.3, 4.3 and 3.1) have preferential expression in embryo, but weak in other tissues. Consequently, the expression level of individual OsAP2-3.5 is higher in shoot development and post-emergence inflorescence (Post EI) as compared to the pre-emergence inflorescence (Pre EI) (Fig. 5). Relative expression pattern of AP2 genes in Chinese cabbage revealed that BrAP2 transcription factor transcripts are most abundant in roots (31.95%) followed by seeds (22.90%), flowers (13.14%) and leaves (20.06%) [20]. Additionally, the expression pattern of cucumber AP2/ERF revealed that among 41 AP2/ERF genes, 35 transcripts expressed in flowers, three (CsERF026, CsERF036, and CsERF045) in leaves, two (CsERF067 and CsERF072) in roots and one (CsERF067) in shoots [90]. Recent studies on *Brachypodium distachyon* AP2 TFs (BdAP2) exhibited that BdAP2/ERF005 and BdAP2/ERF006 have an abundant expression in the developing inflorescence. Similarly, BdAP2/ERF092, BdAP2/ERF131, BdAP2/ERF011, BdAP2/ERF012, BdAP2/ERF013 and BdAP2/ERF139 were recorded to be chiefly expressed during pollination, which further contributes to the seed development in *B. distachyon* [91].

3.11. Expression analysis of OsAP2 genes under heat stress

AP2 transcription factors are known to play important regulatory functions in abiotic stresses such as heat, cold, salinity and drought

tolerance [2,92–94]. High temperature is a serious environmental stress that causes heavy yield reductions globally [95]. However, role of most of the AP2 genes in heat stress response has not been studied yet. Thus, we analyzed the role of AP2 subfamily genes in regulating heat stress tolerance using real-time expression analysis. We performed the qRT-PCR for all 26 AP2 subfamily genes using contrasting heat tolerant *indica* rice genotypes HTT-112 (heat tolerant) and HTT-1 (heat sensitive) reported previously [43]. Our results divided the AP2 genes into three categories; (1) up-regulated genes, (2) down-regulated genes, and (3) the genes with no considerable change in gene expression under heat stress (Fig. 6). The genes OsAP2-4.1, OsAP2-5.3, OsAP2-6.1, OsAP2-6.2, OsAP2-6.4, and OsAP2-8.1 were up-regulated under heat stress in the heat tolerant genotype HTT-112 (Fig. 6), suggesting their important role in regulating heat tolerance in rice.

Leaf trichomes are specialized structures on the leaf surface in various plants, that play crucial roles in resisting external environmental stress including heat stress by moderation of leaf temperature [96]. Recently, an OsAP2 sub-family member has been identified to take active part in leaf trichome initiation and then elongation [97,98]. OsAP2-6.4 also known as OsGL6 (GLABRA6) (i.e. LOC_Os06g44750), when over-expressed has been shown to stimulate and promote trichome initiation in rice leaves [98]. Its allele HL6 (Hairy leaf 6) in other rice genotype has also been involved in leaf trichome elongation [97]. During this process, HL6 physically interacts with OsWOX3B to

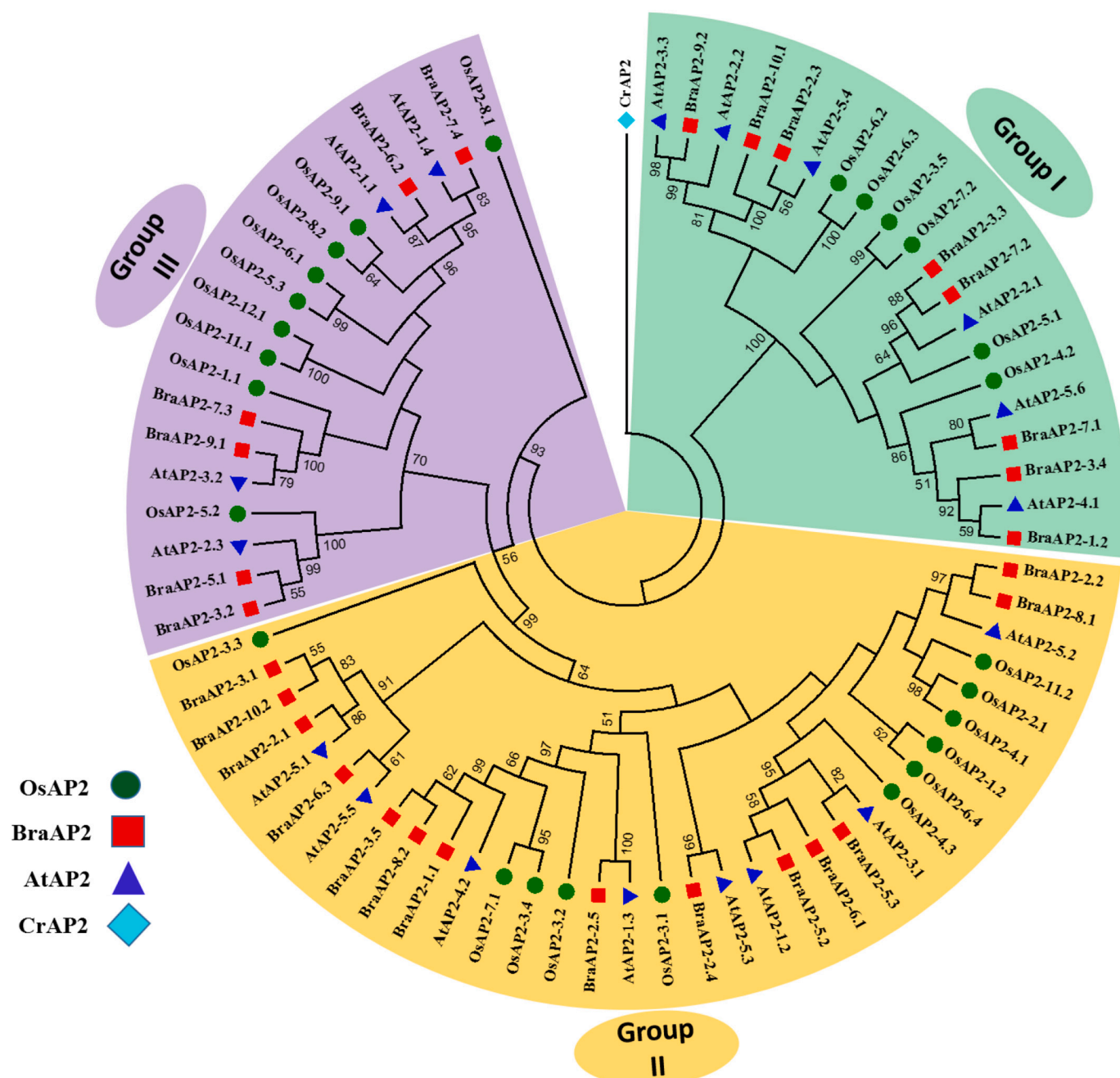


Fig. 7. Comparative phylogenetic tree of AP2 subfamily members of *Oryza sativa*, *Brassica rapa*, *Arabidopsis thaliana* and *Chlamydomonas*. The tree clearly divides the AP2s in three distinct groups. It is also evident from the tree that AP2s from *B. rapa* are closer to AP2s of *A. thaliana* as compared to *O. sativa*. Members of Group I share more similarity with the ancient AP2 of *Chlamydomonas* (CrAP2).

regulate the expression of auxin-related genes which in turn coordinate trichome elongation.

The genes *OsAP2-2.1*, *OsAP2-3.2*, *OsAP2-3.3* and *OsAP2-9.1* were down-regulated under heat stress in the heat tolerant cultivar HTT-112 (Fig. 6), which indicated that these genes may negatively regulate heat tolerance in rice or it can be a compensatory response due to heat stress [99]. Quite recently, a comparative transcriptome analysis has been carried out to identify the stress-specific and cultivar-specific genes under heat stress conditions in two rice genotypes contrasting in heat tolerance [100]. Among other genes, *OsAP2-3.3* (*BG10SGA010867*) has also been shown to differentially express during panicle development under heat stress conditions in heat-resistant rice cultivar. On contrary, *OsAP2-6.2* (*BG10SGA020813*) and *OsAP2-7.1* (*BG10SGA025077*) have

been demonstrated as heat stress-responsive genes in both heat-susceptible and heat-resistant cultivars [100]. Thus, it can be concluded that some *OsAP2* genes show their expression in stress and tissue specific manner, while other show expression in a stress-specific manner.

The genes *OsAP2-1.1*, *OsAP2-1.2*, *OsAP2-3.4*, *OsAP2-3.5*, *OsAP2-4.3*, *OsAP2-5.1*, *OsAP2-5.2*, *OsAP2-6.3*, *OsAP2-7.2*, *OsAP2-8.2*, *OsAP2-11.1* and *OsAP2-12.1* showed no considerable change in expression under heat stress in both genotypes, suggesting that these genes may be involved in regulating other abiotic stresses rather than heat stress. A recent study on comparative analysis of salinity responsive network between *Arabidopsis* and rice revealed the involvement of development and growth-related AP2 genes in rice for mediating salinity responses [93]. Among other

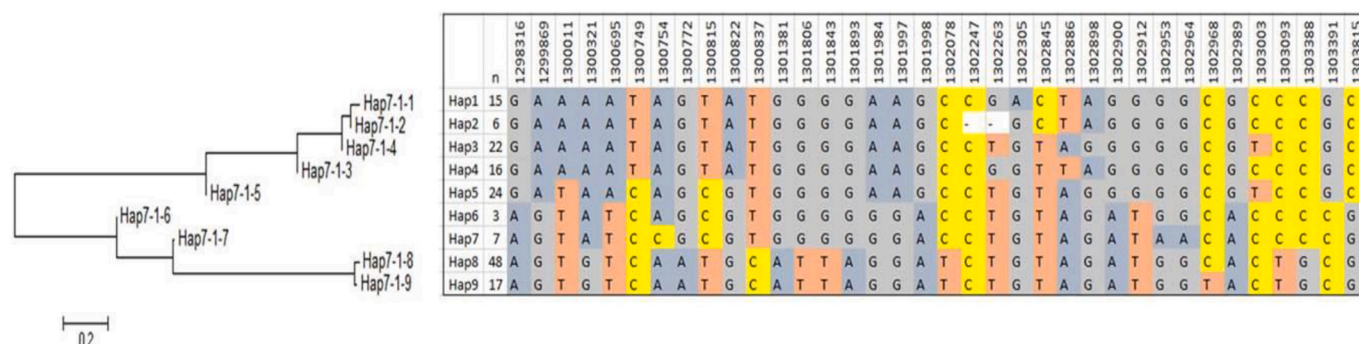


Fig. 8. Haplotype study and phylogenetic tree for *OsAP2-7.1*. The positions of amino acids are shown at the upper side of figure. The different haplotypes along with their phylogenetic relationship are shown on the left side.

genes, *OsAP2-3.4* (*Os03t0770700*) and *OsAP2-3.5* (*Os03t0818800*) has been shown to take part in avoiding salt stress in rice by regulating growth and development [93]. On contrary, more stress directly-related genes rather than growth- and development-related genes were detected in *Arabidopsis* networks under salinity stress.

We have not detected any expression at four AP2 genes namely *OsAP2-3.1*, *OsAP2-4.2*, *OsAP2-7.1* and *OsAP2-11.2* which could be due to the reason that these genes might have different spatio-temporal expression pattern. The tissue specific expression profile reveal that the *OsAP2-3.1* and *-7.1* gene were highly expressed in embryo, and *OsAP2-4.2* in shoot pistil and endosperm while *OsAP2-11.2* did not expression (<0.25 inter quartile value) (Fig. 5). Nonetheless, all of these four genes had very low expression in leaves (Fig. 5) and did not expressed in heat stress. It may due to their tissue or genotype specific expression pattern which could not identify in stress conditions [93,100].

In short, variations in expression patterns were observed in this study among *OsAP2* genes belonging to all the three groups. It has been proposed that the variations in the cis-regulatory elements of AP2 genes could be correlated with lineage-specific expression divergence in rice [101]. Previously, response of AP2 family genes in various abiotic stresses such as drought [99], heat and cold [102], reproductive development [103] and plant growth [104] has been documented. Taken together, our expression data suggest that role of AP2 subfamily in heat stress tolerance does not depend on a particular class but depends on individual genes. Further, functional characterization of these key genes (*OsAP2-4.1*, *OsAP2-5.3*, *OsAP2-6.1*, *OsAP2-6.2*, *OsAP2-6.4*, *OsAP2-6.5*, *OsAP2-8.1*) using genome editing will help to validate their role and study the mechanism of these genes in heat tolerance [105].

3.12. Comparative phylogenetic and evolutionary relationships of AP2 members from *Oryza sativa*, *Brassica rapa*, *Arabidopsis thaliana*, and *Chlamydomonas*

To compare the AP2 protein from various species, full-length protein sequences of 26 AP2 proteins from *Oryza sativa* L. ssp. *indica*, 18 AP2s from *Arabidopsis*, 28 AP2s from *Brassica rapa* were used to generate a comparative phylogenetic tree. The tree was then rooted using the only AP2 from green algae *Chlamydomonas* (*CrAP2*) which is believed to be the ancestor of the AP2 family of higher plants.

The resultant phylogenetic tree divided the proteins into three major clades i.e. group I, II and III, strongly supported by higher bootstrap values (Fig. 7). The AP2 proteins from all the aforesaid species were represented in these three groups. A total of 13 orthologous and 13 paralogous pairs were identified in the tree. Interestingly it was noticed that *Arabidopsis* AP2 proteins are preferentially present in the orthologous relationship with AP2 proteins of *B. rapa*. All the 13 orthologous relationships in the tree were observed from AtAP2 and BrAP2 proteins. These results are in accordance with the fact that the most similar

proteins observed in the phylogenetic tree tended to originate from closely related species [106]. Moreover, the co-orthologous relationship of various rice AP2 proteins with rapeseed and *Arabidopsis* AP2s were also observed. Most of the orthologous AtAP2s have been functionally characterized suggesting the possibility of rice co-orthologs to have the same function, as orthologous genes are generally assumed to retain a high level of functional conservation in different organisms [107].

Group I represent a monophyletic group consisting of 20 AP2 proteins; 8 of them were BrAP2, 6 AtAP2 and 6 OsAP2 (Fig. 7). In this group, a total of seven homologous gene pairs were observed; four OGPs and three PGPs. Among them, AtAP2-3.3 (AT3G54990), also known as SCHLAFMUTZE (SMZ) from *Arabidopsis*, is a co-ortholog of *OsAP2-6.2* and *OsAP2-6.3* which may indicate their co-evolution. The SMZ gene was stated to be intricate in the repression of flowering when targeted by microRNA miR172 [17]. In *Arabidopsis*, floral (organ identity, establishment of floral meristem identity and destruction of floral meristem) and seed characteristics (ovule and seed coat development) are regulated by another protein i.e. AtAP2-4.1 (AT4G36920) [108–111]. Being an ortholog of AtAP2-4, *OsAP2-4.2* might be responsible for floral organ identity. The AtAP2-5.6 (AT5G67180) also known as TOE3, a co-ortholog of *OsAP2-4.2* gene. The TOE3 gene suppresses AGAMOUS expression during *Arabidopsis* floral patterning [77].

In Group II, 33 AP2 proteins were present. Among them, segmentally duplicated paralogous genes *OsAP2-3.4* and *OsAP2-7.1* are co-orthologous of *Arabidopsis* gene “AtAP2-4.2” which is also known as INTEGUMENTA (ANT). The ANT protein was reported to be responsible in many functions of growths and development e.g. floral growth and patterning [112–114] and plant organ size control [115].

The *Arabidopsis* TF AtAP2-3.2 (from Group III), also known as WRINKLED1, is orthologous of *OsAP2-1.1*. WRINKLED1 has been associated in regulation of many genes involved in diverse functions e.g. biosynthesis of storage compounds during seed formation [11], glycolytic and fatty acid biosynthesis [14,15] and plant oil creation in seed and non-seed tissues [15]. Since *OsAP2-1.1* is a close ortholog of WRINKLED1 gene, thus it can be predicted that *OsAP2-1.1* may also be involved in these functions in *Oryza sativa* L. subsp. *indica*.

The *Arabidopsis* TF AtAP2-01 also called as ARIA-interacting double AP2 domain protein (ADAP), is an ortholog of *OsAP2-8.2* and *OsAP2-9.1*. Absciscic acid (ABA) is positively regulated by ADAP and it is also responsible for the regulation of germination and seedling developmental stages [16]. Thus, it can be suggested that *OsAP2-8.2* and *OsAP2-9.1* may also be involved in ABA signaling and regulation of seed germination and growth.

3.13. Haplotype analysis of *OsAP2* genes

The different haplotypes of AP2 loci were constructed after removal of singleton and monomorphic variable sites and SNPs having less than 10% minor allele frequency. Initially, higher number of haplotypes was

created with maximum variable sites. The rare haplotypes with less than three entries were exempted. Finally, a minimum of two haplotypes for *OsAP2-4.1* based on twenty SNPs to maximum nine haplotypes for locus *OsAP2-7.1* with 36 SNPs were constructed. The overall haplotype diversity ranged from 0.517 for *OsAP2-4.1* to 0.847 for *OsAP2-7.1* (Table S5).

For *OsAP2-7.1* two clear clades can be observed (Fig. 8) which indicated the Hap5 and Hap6 as the wild ancestor. Four other variations (Hap-1 to Hap-4) occurred during the process of domestication. Among the members of clade-I, the Hap-5 was widely accepted and evolved by continuous selection, followed by Hap-3. It further revealed that the wild allele was selected during domestication. Inversely in clade-II, the ancestral haplotype Hap-6 was divided into three haplotypes (Hap-7, Hap-8, Hap9) while Hap-8 was preferred during natural or artificial selection. In a phylogenetic study of haplotypes for all the loci, the similar trend was observed. The selected loci revealed a clear divergence into two major clades. In each clade, one of the haplotypes was observed as main and most diverse while others were derived from that haplotype. The other haplotypes may result from mutation and domesticated by natural or artificial selection.

According to various evolutionary studies of rice, it has been known that *indica* and *japonica* rice may evolve from two diverse wild species or from a common ancestor domesticated at two different ecosystems [116]. Similarly, in our study, the two clades division may have an indication of *indica* domestication at various places.

4. Conclusion

In the present study, we have identified 26 AP2 genes. These AP2 genes were found to be involved in various developmental and stress-response mechanisms. The AP2 genes play a major role in the seed development and germination as they showed high expression in seed and flower tissues. Since only a few AP2 genes in rice were characterized yet, this is the first comprehensive study of this AP2 TF subfamily. Furthermore, the comparative study of these AP2 genes with model plant *Arabidopsis* enabled us to predict their involvement in the different development and stress-related pathways and its protein interaction. This provides us with a strong base for molecular characterization of AP2 genes in rice, especially under stress conditions.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jgeno.2020.10.037>.

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Declaration of Competing Interest

The research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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